MCP Style Interface Specification
SME Procurement-Preparation Agent
GROUP J EVENING
BSE4104
Week 6 Deliverable

1. Purpose
   This document specifies one project capability using an MCP-style (Model Context Protocol style) interface.
   It defines the capability clearly so that other agents, tools, or systems can interact with it in a controlled and secure way.

   Chosen Capability
   create_draft_requisition

2. Capability Overview

Field	Value
Capability Name	create_draft_requisition
Display Name	Create Draft Purchase Requisition
Version	1.0
Type	Action
Side effect	Yes
Idemponent	No
Requires confirmation	No
Human approval required for final action	Yes

3. Description
   Creates a new Purchase Requisition in “Draft – Awaiting Approval” status.
   The capability allows an agent or authorised client to prepare a requisition.

It does not approve the requisition, select a final supplier, place an order, or commit funds.

4. Input Schema

```
{
  "type": "object",
  "properties": {
    "item_id": {
      "type": "string",
      "description": "Unique inventory item identifier (e.g. ITM-001)"
    },
    "quantity": {
      "type": "number",
      "minimum": 1,
      "description": "Quantity to request. Must be a positive number."
    },
    "justification": {
      "type": "string",
      "description": "Reason for creating the requisition"
    },
    "requested_by": {
      "type": "string",
      "description": "Name or ID of the person or system requesting the draft"
    }
  },
  "required": ["item_id", "quantity", "justification", "requested_by"],
  "additionalProperties": false
}
```

5. Output Schema

```
{
  "type": "object",
  "properties": {
    "requisition_id": {
      "type": "string",
      "description": "Unique identifier of the created draft (e.g. PR-A1B2C3D4)"
    },
    "item_id": { "type": "string" },
    "quantity": { "type": "number" },
    "status": {
      "type": "string",
      "enum": ["Draft - Awaiting Approval"]
    },
    "created_at": {
      "type": "string",
      "format": "date-time"
    },
    "message": {
      "type": "string"
    }
  },
  "required": ["requisition_id", "item_id", "quantity", "status", "created_at"]
}
```

6. Error Responses

Error code	When it occurs	Example message
ValidationError	Missing or invalid parameters	"quantity must be a number >= 1"
NotFound	item_id does not exist in inventory	"Item 'ITM-999' not found in inventory"
Unauthorized	Caller lacks permission	"Caller is not authorised to create requisitions"
ToolExecutionError	Unexpected internal failure	"Failed to save draft requisition"

7. Permissions & Security Boundary

Aspect	Rule
Who can call this capability	Authenticated procurement users or the bounded agent only
What it is allowed to do	Create a record with status “Draft – Awaiting Approval” only
What it is forbidden to do	Change status to Approved, place orders, select final suppliers, commit funds
Data access	May read inventory data to validate item_id. May write only to the requisitions store
Audit	Every call is logged with timestamp, caller, and parameters
Human control	Final approval remains outside this capability and requires explicit human action

8. Example Call
Request

```
{
  "item_id": "ITM-001",
  "quantity": 50,
  "justification": "Stock below reorder point",
  "requested_by": "Jane Auma"
}
```

9. Successful Response

```
{
  "requisition_id": "PR-A1B2C3D4",
  "item_id": "ITM-001",
  "quantity": 50,
  "status": "Draft - Awaiting Approval",
  "created_at": "2026-10-08T14:22:10",
  "message": "Draft requisition created successfully. Awaiting human approval."
}
```

10. Integration
    - This capability can be exposed to other agents or systems via a simple function calling or HTTP interface.
    - The interface follows the principle of least privilege: it only performs the minimum necessary side effect.
    - Callers should treat the returned requisition_id as a handle for later human review.
    - The capability is deliberately non-idempotent; repeated calls create new draft records.